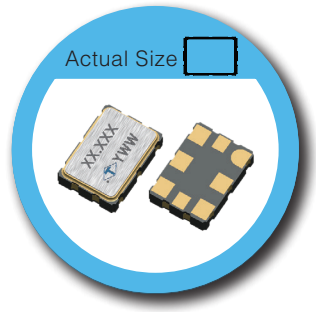
OD-P Type High Frequency, Ultra Low Noise and Frequency Selectable 7.0 x 5.0mm Quad Frequency SMD Crystal Oscillator



RoHS Compliant

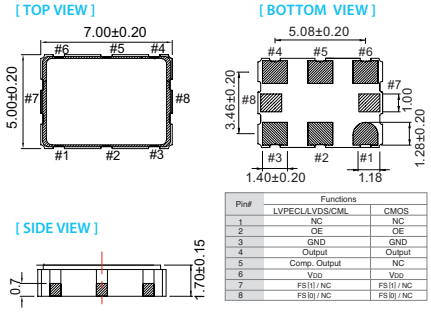
FEATURE

- Any Frequency Quad Frequency Selectable.
- Low Power Supply Voltage: 3.3, 2.5 and 1.8V Supply Options
- Clock Output: LVPECL, LVDS, CML, HCSL and LVCMOS
- Output Frequency Support from 15MHz to 2.1GHz
- Ultra Low Noise, Phase Jitter < 300 fs
  (Typical: 150 fs at 12kHz to 20MHz Frequency Offsets)
- Tri-state Enable / Disable Mode.
- Temperature Range: -40 to 85°C
- Pb-free/RoHS Compliant

TYPICAL APPLICATION

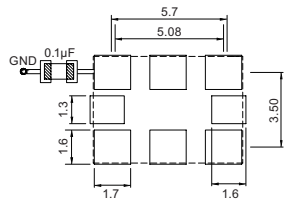
- SONET/SDH, Gigabit Ethernet
- Storage Area Networking (SAN)
- SD/HD Video
- FPGA Clock Generation

DIMENSION (mm)



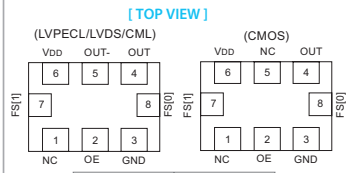
Pin#	Functions	
	LVPECL/LVDS/CML	CMOS
1	NC	NC
2	OE	OE
3	GND	GND
4	Output	Output
5	Comp. Output	NC
6	VDD	VDD
7	FS[1] / NC	FS[1] / NC
8	FS[0] / NC	FS[0] / NC

SOLDER PAD LAYOUT(mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between Vdd and GND pads.

PIN ASSIGNMENTS



Select Pins		Output
FS[1]	FS[0]	
0	0	Frequency 0
0	1	Frequency 1
1	0	Frequency 2
1	1	Frequency 3

0:Logic Low, Contact to GND or Voltage < 10%VDD
1:Logic High, VDD, Contact to VDD or Voltage > 90%VDD

ELECTRICAL SPECIFICATION

Parameter			LVPECL 3.3V Min.	LVPECL 3.3V Max.	LVPECL 2.5V Min.	LVPECL 2.5V Max.	unit
Supply Voltage Variation (VDD) ±10%			VDD-10%	VDD+10%	VDD-10%	VDD+10%	V
Frequency Range			15	2100	15	2100	MHz
Standard Frequency			155.52, 156.25, 187.5, 212.5,250, 312.5, 622.08, 805.664, 873.515MHz				MHz
Supply Current			–	110	–	95	mA
Output Level		Output High	VDD-1.165	VDD-0.8	VDD-1.165	VDD-0.8	V
		Output Low	VDD-2.0	VDD-1.55	VDD-2.0	VDD-1.55	V
Transition Time (20%-80%)	Rise Time / Fall Time		–	0.35	–	0.35	nSec
Duty Cycle			45	55	45	55	%
Startup Time			–	8	–	8	mSec
Tri-State Mode (Input to Pin 2)		Enable	0.7-VDD	–	0.7 x VDD	–	V
		Disable	–	0.3 x VDD	–	0.3 x VDD	V
Standby Current			–	110	–	95	mA
Phase Noise			Typ.	Max.	Typ.	Max.	
At VDD=3.3V ,Fout=873.515MHz		1kHz offset	-106	–	-106	–	dBc/Hz
		10kHz offset	-115	–	-115	–	dBc/Hz
		100kHz offset	-123	–	-123	–	dBc/Hz
		1MHz offset	-133	–	-133	–	dBc/Hz
		20MHz offset	-150	–	-150	–	dBc/Hz
RMS Phase Jitter (12kHz to 20MHz)			150	300	150	300	fs
Period Jitter			–	50	–	50	ps

Note: not all combination of options are available. Other specifications may be available upon request.

Specifications subject to change without notice.

www.taitien.com
sales@taitien.com.tw

Rev(7) 05/2020

Parameter			LVDS						unit
			3.3V		2.5V		1.8V		
			Min.	Max.	Min.	Max.	Min.	Max.	
Supply Voltage Variation (VDD) ±5%			–	–	–	–	1.71	1.89	V
Supply Voltage Variation (VDD) ±10%			3.63	2.97	2.25	2.75	–	–	
Frequency Range			15	2100	15	2100	15	2100	MHz
Standard Frequency			155.52, 156.25, 187.5, 212.5, 312.5, 622.08, 805.664, 873.515MHz						
Supply Current			–	90	–	80	–	70	mA
Output Level	Output High		–	1.6	–	1.6	–	1.6	V
	Output Low		0.9	–	0.9	–	0.9	–	
Transition Time (20%-80%)	Rise Time / Fall Time		–	0.35	–	0.35	–	0.35	nSec
Duty Cycle			45	55	45	55	45	55	%
Startup Time			–	8	–	8	–	8	mSec
Tri-State Mode (Input to Pin 2)		Enable	0.7 x VDD	–	0.7 x VDD	–	0.7 x VDD	–	V
		Disable	–	0.3 x VDD	–	0.3 x VDD	–	0.3 x VDD	
Standby Current			–	90	–	80	–	70	mA
Phase Noise			Typ.	Max.	Typ.	Max.	Typ.	Max.	
At VDD=3.3V, Fout=873.515MHz		1kHz offset	-106	–	-106	–	-106	–	dBc/Hz
		10kHz offset	-115	–	-115	–	-115	–	
		100kHz offset	-123	–	-123	–	-123	–	
		1MHz offset	-133	–	-133	–	-133	–	
		10MHz offset	-150	–	-150	–	-150	–	
RMS Phase Jitter (12kHz to 20MHz)			150	300	150	300	150	300	fs
Period Jitter			–	50	–	50	–	50	ps

Parameter			CML						unit
			3.3V		2.5V		1.8V		
			Min.	Max.	Min.	Max.	Min.	Max.	
Supply Voltage Variation (VDD) ±5%			–	–	–	–	1.71	1.89	V
Supply Voltage Variation (VDD) ±10%			3.63	2.97	2.25	2.75	–	–	
Frequency Range			15	2100	15	2100	15	2100	MHz
Standard Frequency			155.52, 156.25, 187.5, 212.5, 312.5, 622.08, 805.664, 873.515MHz						
Supply Current			–	90	–	80	–	70	mA
Output Level	Output High		VDD-0.085	VDD	VDD-0.085	VDD	VDD-0.085	VDD	V
	Output Low		VDD-0.6	VDD-0.32	VDD-0.6	VDD-0.32	VDD-0.6	VDD-0.32	
Transition Time (20%-80%)	Rise Time / Fall Time		–	0.35	–	0.35	–	0.35	nSec
Duty Cycle			45	55	45	55	45	55	%
Startup Time			–	8	–	8	–	8	mSec
Tri-State Mode (Input to Pin 2)		Enable	0.7 x VDD	–	0.7 x VDD	–	0.7 x VDD	–	V
		Disable	–	0.3 x VDD	–	0.3 x VDD	–	0.3 x VDD	
Standby Current			–	90	–	80	–	70	mA
Phase Noise			Typ.	Max.	Typ.	Max.	Typ.	Max.	
At VDD=3.3V, Fout=873.515MHz		1kHz offset	-107	–	-107	–	-107	–	dBc/Hz
		10kHz offset	-117	–	-117	–	-117	–	
		100kHz offset	-125	–	-125	–	-125	–	
		1MHz offset	-135	–	-135	–	-135	–	
		10MHz offset	-150	–	-150	–	-150	–	
RMS Phase Jitter (12kHz to 20MHz)			150	300	150	300	150	300	fs
Period Jitter			–	50	–	50	–	50	ps

Parameter		HCSL						Unit
		3.3V		2.5V		1.8V		
		Min.	Max.	Min	Max.	Min.	Max.	
Supply Voltage Variation (VDD)±10%		3.63	2.97	2.25	2.75	1.71	1.89	V
Frequency Range		15	700	15	700	15	700	MHz
Supply Current		-	115	-	100	-	94	mA
Output Level	Output High	0.66	1.15	0.66	1.15	0.66	1.15	V
	Output Low	0	0.15	0	0.15	0	0.15	V
Transition Time (20% - 80%)	Rise Time	-	0.4	-	0.4	-	0.4	nSec
	Fall Time	-	0.4	-	0.4	-	0.4	nSec
Duty Cycle		45	55		55	45	55	%
Startup Time		-	8		8	-	8	mSec
Tri-State mode (Input to Pin 2)	Enable	0.7 x VDD	-	0.7x VDD	-	0.7xVDD	-	V
	Disable	-	0.3 x VDD	-	0.3 x VDD	-	0.3 x VDD	V
Stand by Current	Disable	-	115	-	100	-	94	mA
Output Load		50 ohms to GND						
Phase Noise		Typ.	Max.	Typ.	Max.	Typ.	Max.	
At VDD=3.3V, fout= 873.515MHz	1kHz offset	-87	-	-87	-	-87	-	dBc/Hz
	10kHz offset	-110	-	-110	-	-110	-	dBc/Hz
	100kHz offset	-127	-	-127	-	-127	-	dBc/Hz
	1MHz offset	-138	-	-138	-	-138	-	dBc/Hz
	10MHz offset	-153	-	-153	-	-153	-	dBc/Hz
RMS Phase Jitter (12KHz to 20MHz)		150	300	150	300	150	300	fs
Period Jitter		-	50	-	50	-	50	ps

Note: not all combination of options are available. Other specifications may be available upon request.

Parameter			CMOS						unit
			3.3V		2.5V		1.8V		
			Min.	Max.	Min.	Max.	Min.	Max.	
Supply Voltage Variation (V_{DD}) ±5%			–	–	–	–	1.71	1.89	V
Supply Voltage Variation (V_{DD}) ±10%			3.63	2.97	2.25	2.75	–	–	
Frequency Range			15	250	15	250	15	250	MHz
Standard Frequency			106.25, 125, 133.33, 150, 155.52, 156.25, 187.5, 212.5MHz						
Supply Current			–	90		80		70	mA
Output Level	Output High		0.9 X V_{DD}	–	0.9 X V_{DD}	–	0.9 X V_{DD}	–	V
	Output Low		–	0.1X V_{DD}	–	0.1X V_{DD}	–	0.1X V_{DD}	
Transition Time (20%-80%)	Rise Time / Fall Time		–	1.2	–	1.5	–	2	nSec
Duty Cycle	Fout <100MHz		45	55	45	55	45	55	%
	Fout >100MHz		40	60	40	60	40	60	
Startup Time			–	8	–	8	–	8	mSec
Tri-State Mode (Input to Pin 2)		Enable	0.7 x V_{DD}	–	0.7 x V_{DD}	–	0.7 x V_{DD}	–	V
		Disable	–	0.3 x V_{DD}	–	0.3 x V_{DD}	–	0.3 x V_{DD}	
Period Jitter			–	100	–	100	–	100	mA

FREQ. STABILITY vs. TEMP. RANGE

Temp. (°C) \ ppm	±20	±25	±30	±50
-20~+70	△	○	○	○
-40~+85	X	△	○	○

* ○: Available △:Conditional X: Not available
* Inclusive of calibration @ 25 °C, operating temperature range, input voltage variation, load variation, aging (1st year), shock, and vibration